

Measuring the Efficacy of a Low-Cost Tendon-Driven Prosthetic Arm

Evaluating the dexterity, strength, and durability of prosthetics made with affordable materials

Dr. Yolanda Sheppard · Dr. Yolanda Sheppard · **Dr. Yolanda Sheppard**
STEM Center · Bergen Community College · Pathways Grant

The Problem

The most advanced commercial hands offer select few preset grip patterns at a price out of reach for most users, and those who do receive them stop wearing them, citing limited function, excessive weight, and overall dissatisfaction.

44%

Salminger et al., 2022

of upper-limb amputees reject their prosthesis. Acceptance showed no significant difference before vs. after 2006.

81%

Yamamoto et al., 2019

of amputees who do not use a prosthesis named limited functionality as their reason for abandoning the device.

\$6,000

Open Bionics, 2026

Lowest pricing of the hero arm, known to be one of the cheaper myoelectric prosthetics available.

Our Mission

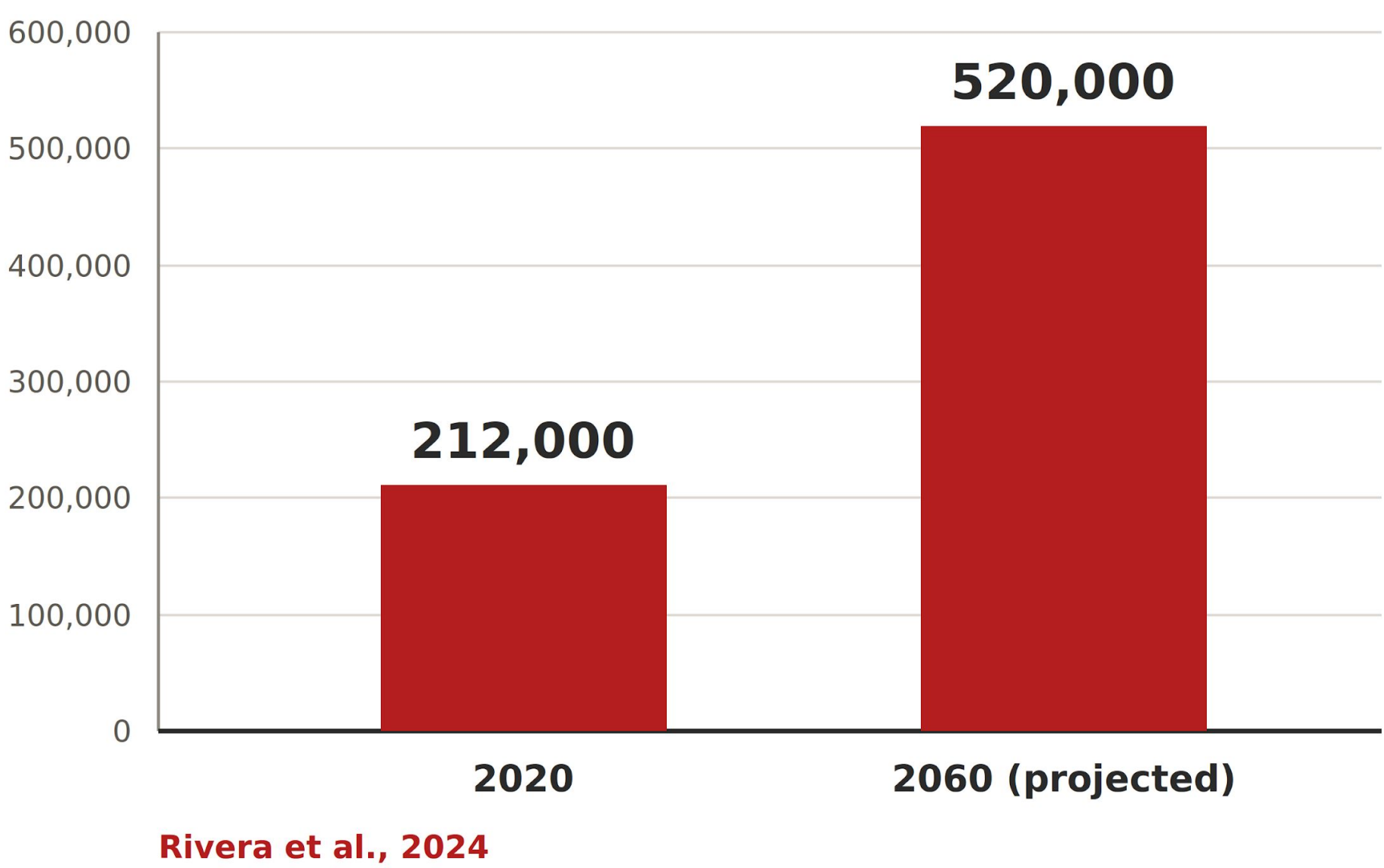
Develop a tendon-driven anthropomorphic robotic prosthetic arm with an accessible cost while ensuring full functionality and improved quality of life for users who lack access to more expensive proprietary options.

Affordable Cheap servos, 3D printed structure, accessible materials.

Dexterous Tested against 33 gestures of Feix GRASP taxonomy.

Reproducible Open source CAD, firmware, and build documentation

Projected US upper-limb loss, 2020-2060



Our Proposal — The System

“ARMin” is a proof-of-concept prosthetic arm driven by hobby-grade servos through fishing-line tendons, and stepper motor driven robotic actuators, with all control on board a single microcontroller and an off-board vision path for teleoperation.



\$290.54
Cost per finished unit

Actuation

6 Servos, 5 Micro Servos, 2 Stepper Motors.

Hand

Fishing-line tendons with elastic return and direct-drive abduction joints.

2 DOF Elbow

2 Planetary Robotic Actuators of a 64:1 reduction and 4:1 reduction controlling two degrees of freedom at the elbow joint.

Control

Raspberry Pi uses Computer Vision to characterize gestures, passing the data to the ESP32, which controls the arm.

Structure

Fully 3D Printed Structure, utilizing a combination of PLA and PETG-CF.

Power

4S 3700mAh LiPo → 30 A buck converter → Terminal Block → PCA9685 PWM driver

Hypothesis

Hypothesis:

A prosthetic arm made for under \$300 will achieve at least 18 of the 33 GRASP taxonomy grasp types with secure object retention, along with a torque output of at least 10 N · m at the elbow.

Alternative:

A prosthetic arm made for under \$300 will achieve at least 18 of the 33 GRASP taxonomy grasp types or generate a torque output of at least 10 N · m at the elbow, but not both.

Null:

A prosthetic arm made for under \$300 will not achieve at least 18 of the 33 GRASP taxonomy grasp types, nor will it generate a torque output of at least 10 N · m at the elbow.

Evaluation Methods

Measurement	Procedure and Implications
GRASP taxonomy assessment	All 33 Grasps attempted with fixed testing environment, generating a resultant score measuring the dexterity and articulation capabilities of the hand.
Elbow payload	Torque measured at the elbow (N-m) through a horizontal to vertical lift, which is tested with a calibrated mass set.
Tendon durability	Completion of 10,000 open-close cycles or to first failure, determining the material durability, which will be measured using a load cell to track tension in tendons.

Next Steps

Equipment request

Pull gauge, calibrated mass set, load cell, cycling rig.

Data Collection

Develop a profile of our arm compared to industry standards in our benchmarks

Beyond this grant

Fully open-source the project, Brain Computer Interface, Mounting Harness

Supporting Data

Device	Selectable grips	Approximate user cost
i-LIMB Pulse (2011 generation)	10 pre-programmed grip patterns	\$18,000
bebionic v2 (2011 generation)	11 pre-programmed grip patterns	\$11,000
Open Bionics Hero Arm	4 grip patterns (3 motor version); 6 grip patterns (4 motor version)	\$6,000-\$10,000
ARA “ARMin” (proposed)	Assessed against all 33 GRASP types	\$290.54

Commercial hands are specified by a small set of preset grip patterns, while the GRASP taxonomy supplies an independent standard of 33 types, permitting a universal dexterity benchmark. (Waryck, 2011; Feix et al., 2016; Open Bionics Hero Arm User Manual Version 12.0, 2018)